

Accuracy

1. Overview

Accuracy as a non-functional requirement means:

The system's data, metrics, outputs, and computed results are **correct, precise, and reflect real state**, not stale or approximate.

Accuracy becomes very important in systems involving:

- High write volumes
- Real-time counters
- Analytics pipelines
- Financial systems
- Inventory systems
- Recommendation metrics
- Monitoring dashboards

2. Simple Definition

If system says:

None

Orders today = 12,542

Then true value should actually be 12,542.

If actual is 13,100, then system is inaccurate.

3. Accuracy vs Correctness vs Precision

Term	Meaning
Accuracy	Close to true value
Precision	Repeatable / detailed measurement
Correctness	Business logic valid
Consistency	Replica agreement / latest visibility

4. Why Accuracy Matters

Inaccurate systems cause:

- Wrong balances
- Overselling stock
- Wrong analytics decisions
- Incorrect invoices
- Misleading dashboards
- Fraud detection failures

5. Examples Where Accuracy Is Critical

A. Banking

None

Balance = ₹10,000

Cannot be approximate.

B. Inventory

Last item in stock must be exact.

C. Billing

Usage charges must be accurate.

D. Medical Data

Incorrect dosage or reports dangerous.

6. Examples Where Approximation Is Acceptable

A. Website Visitors Count

None

~2.1M monthly visitors

B. Trending Topics

Approximate ranking okay.

C. Ad Analytics

Small error margin often acceptable.

D. Large Distinct Counts

COUNT DISTINCT at scale.

7. Accuracy vs Latency Trade-off

Exact answers may require heavy computation.

Approximate answers may be faster.

Example:

None

Exact unique users = expensive

Approximate unique users = fast

8. Accuracy vs Cost Trade-off

To improve accuracy, you may need:

- More storage
- More replicas
- More compute
- More synchronization
- Better pipelines
- Slower writes

So perfect accuracy can be expensive.

9. Estimation Algorithms

Used when exact computation is too costly.

A. HyperLogLog

Used for estimating unique counts.

Examples:

- Unique visitors
- Distinct users
- Unique IPs

Memory efficient.

B. Count-Min Sketch

Estimates frequency of events in streams.

Examples:

- Most searched keywords
- Most clicked products
- Top hashtags

10. Example: COUNT DISTINCT

Exact query:

None

```
SELECT COUNT(DISTINCT user_id)
```

On billions rows can be expensive.

Approximate:

Use HyperLogLog.

Fast, tiny memory, small error margin.

11. Streaming Systems Accuracy

Suppose event pipeline counts page views.

Challenges:

- Duplicate events
- Late events
- Lost messages
- Out-of-order events

Need careful design for accurate metrics.

12. Cache Accuracy (Staleness)

Cache becomes inaccurate when DB changed but cache old.

Example:

None

DB price = ₹999

Cache price = ₹899

User sees wrong price.

13. Cache Refresh Approaches

A. TTL / Periodic Refresh

Every 5 minutes refresh cache.

Pros

- Simple

Cons

- Data stale between refreshes

B. Write Invalidate / Update Cache

When DB updates:

None

Update DB

Delete or update cache key

Pros

More accurate

Cons

Higher complexity

14. Accuracy vs Consistency

Related but different.

Consistency

All replicas agree / latest write visible.

Accuracy

Returned data reflects truth.

Example:

All replicas consistently show wrong cached value.

That is:

None

Consistent but inaccurate

15. Eventual Consistency and Temporary Inaccuracy

Write occurs now.

Some replicas still old.

Reads from stale replica give inaccurate result temporarily.

Example:

None

Inventory changed from 1 → 0

Replica still shows 1

Technically we discuss this under **consistency**, but user-facing impact is inaccurate data.

16. Accuracy in High Write Systems

Examples:

- Likes count
- Wallet balance
- Inventory count
- Concurrent edits

Need handling for:

- Race conditions
- Lost updates
- Double counting
- Retry duplication

17. How Accuracy Gets Lost

A. Duplicate Events

Same purchase processed twice.

B. Lost Messages

One event never processed.

C. Race Conditions

Two writers overwrite each other.

D. Stale Reads

Reading old replica.

E. Clock Skew

Wrong event ordering.

18. Techniques to Improve Accuracy

A. Idempotency Keys

Prevent duplicate processing.

B. Transactions

Atomic updates.

C. Strong Consistency

Latest data reads.

D. Deduplication Pipelines

Remove duplicate events.

E. Event Ordering

Use partition keys / sequence numbers.

F. Reconciliation Jobs

Periodic recomputation.

19. Reconciliation / Auditing

Even good systems drift.

Run periodic jobs:

None

Recalculate balances

Recount inventory

Compare source vs derived tables

Correct hidden inaccuracies.

20. Real-World Examples

Payment System

Need exact ledger, no approximation.

Social Media Likes

Approximate counts okay briefly.

Food Delivery ETA

Estimate okay.

Warehouse Inventory

Must be exact.

21. Interview Requirement Questions

Ask:

1. Is exact value required?
2. Acceptable error margin?
3. Real-time or delayed okay?
4. Approximation acceptable?
5. Can stale cache be shown?
6. Financial/legal impact if wrong?

22. Trade-off Table

Choice	Accuracy	Speed	Cost
Exact query	High	Lower	Higher
Approx algorithm	Medium/High	Fast	Low
Strong consistency	High	Slower	Higher
Eventual consistency	Temporary lower	Faster	Lower

23. Interview Answer Example

Q: How would you count unique users on billions of events?

Answer:

If business requires exact billing numbers, I'd compute exact counts offline. If approximate analytics is acceptable, I'd use HyperLogLog for low-memory fast cardinality estimates.

24. Best Practices

- Use exact systems for money/inventory
- Use approximations for analytics at scale
- Define freshness SLAs
- Invalidate cache on writes
- Deduplicate events
- Reconcile periodically
- Measure error rates

25. Short Revision Notes

None

Accuracy = Data reflects true value

Critical:

- Money
- Inventory
- Billing

Approx OK:

- Analytics
- Trends
- Visitor counts

Improve with:

- Transactions
- Idempotency
- Cache invalidation
- Reconciliation